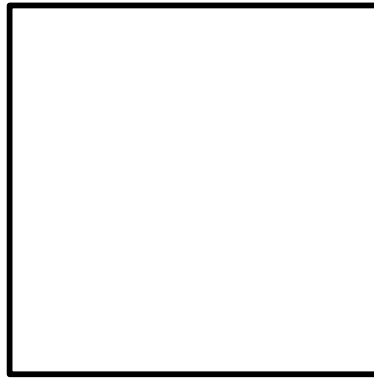


Dissection Puzzle

Ethan Sharma

Welcome to the Dissection Puzzle! For each problem, you will be given a starting figure and a desired figure. Your task is to make a finite number of cuts and rearrange the pieces from the starting figure to the desired figure. For 2-dimensional figures, a cut consists of a line that intersects the figure at two points (you will only be given convex 2-dimensional figures). You are allowed to add/remove segments from the figure as long as the figure remains closed. For a general n -dimensional figure, a cut is a $(n - 1)$ -dimensional hyperplane that cuts the figure into more than one n -dimensional figures. Then, you must put the pieces back together to obtain the desired figure. With this, rearrangement like rotations are allowed. As an example:



(a) Starting Figure (Square)



(b) Desired Figure

Figure 1: Example Dissection

The solution cuts are shown on the next page.

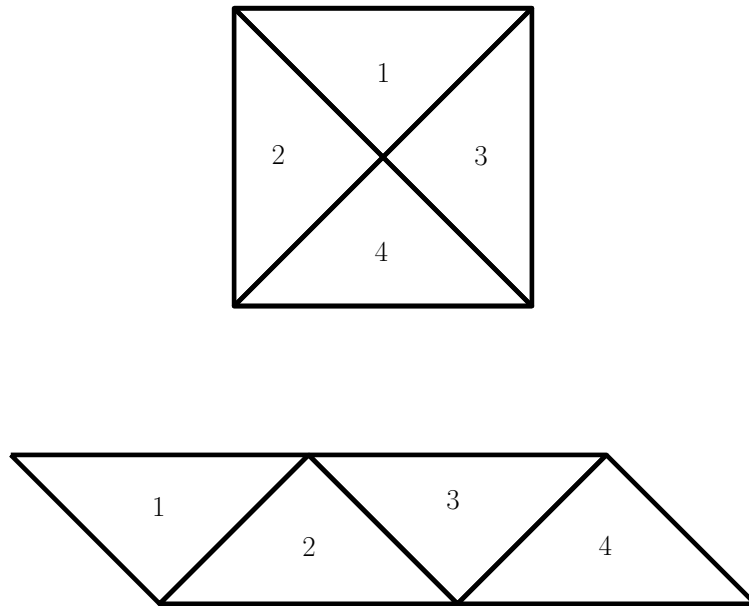


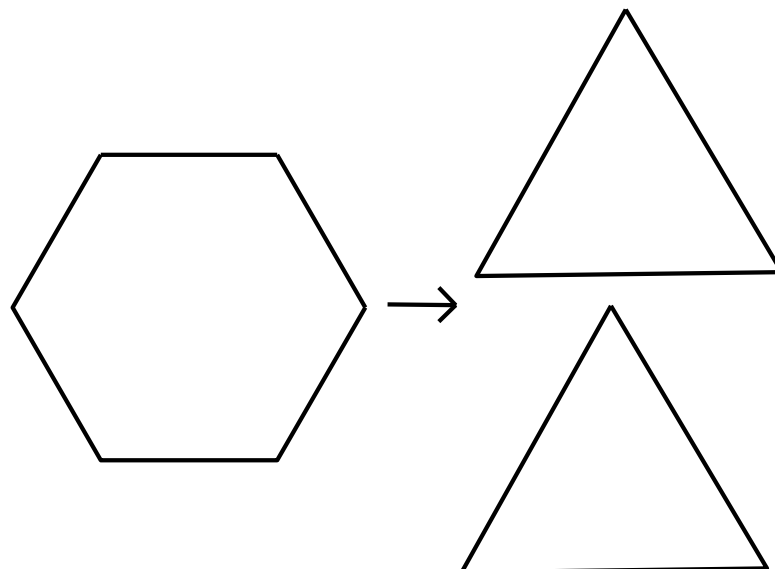
Figure 2: Example Dissection Cut Solution

Note that the order we number the segments here does not matter since they are all congruent.

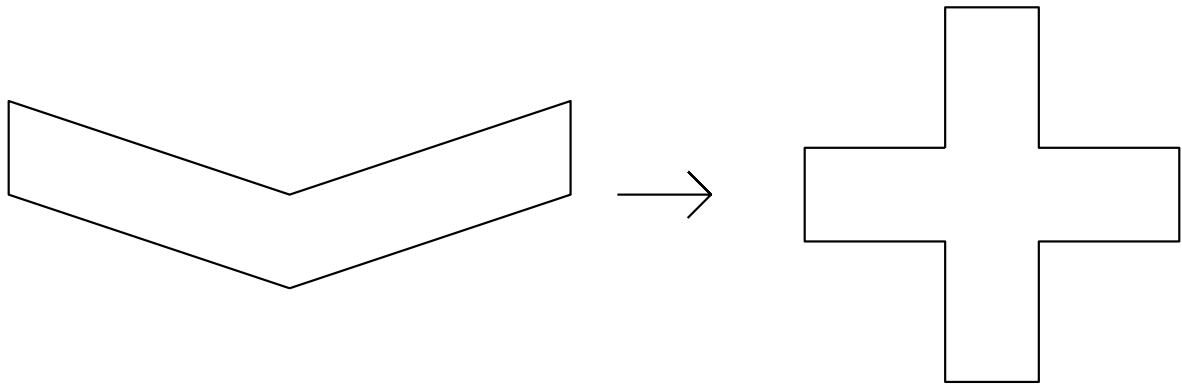
The solution is on page 5.

Problem 1

The starting figure is a regular hexagon, and the desired figure is two congruent equilateral triangles.

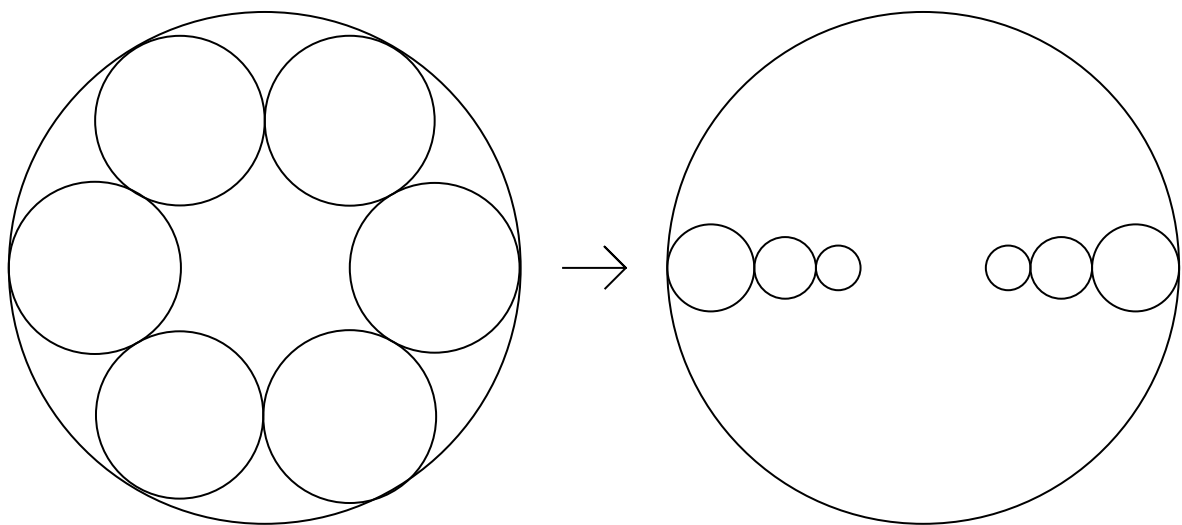


Problem 2



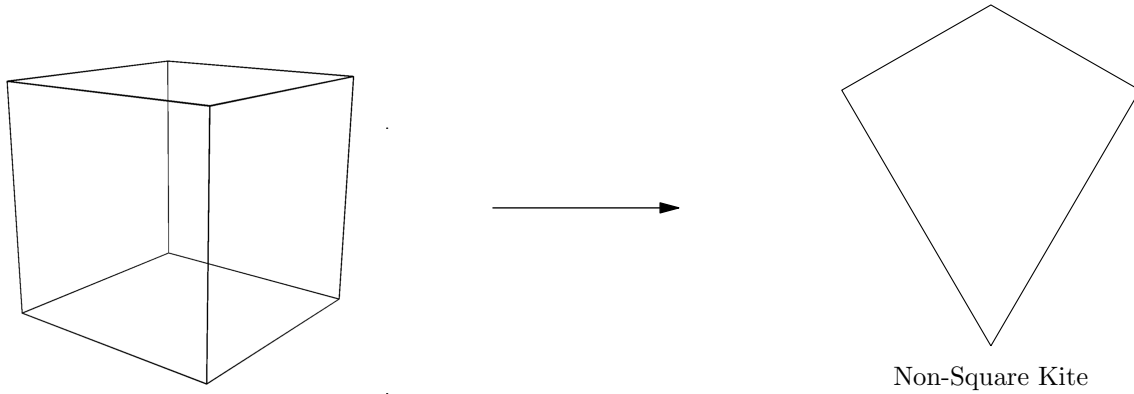
Problem 3

For this problem, you are allowed to perform one geometric inversion, but you must undo it before getting to the final figure (that is, invert again about the same reference circle). The figure isn't directly to scale, but try to get the same configuration.



Problem 4

For this problem, you are allowed to perform an orthogonal projection onto a plane along with cuts and rearrangements. The initial figure is a cube, and the desired figure is a non-square kite.



Problem 5

The initial figure is a cube, and the desired figure shows three cubes of descending side length stacked on each other.

